

A2 FILE CMD

User guide · 0.9.2

ProDOS: 140K essential · 800K complete · XL 6502

XL enhanced 65C02 with optional mouse · DOS3.3

Preparation edition — release qualification in progress

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This guide covers two separate programs:

- **ProDOS:** Apple IIe, //c or IIgs, 128 KB and 80 columns. Choose the essential 140K disk, complete 800K disk, or an XL edition.
- **DOS3.3:** Apple II+ with 48 KB, 40 columns and two Disk II drives. Go directly to A2 File Cmd DOS3.3; ProDOS commands, plugins and recovery filenames do not apply to this separate program.

Start here

The ProDOS **6502** editions run on an original or enhanced IIe, //c or IIgs with 128 KB and 80-column support. XL **65C02** adds MouseText and optional AppleMouse II support, and needs **both** a 65C02 processor and the enhanced ROM: an enhanced IIe, a //c or a IIgs. An enhanced IIe with a 6502 put back in, or an unenhanced IIe with a 65C02 accelerator, needs the 6502 edition. It runs on real Apple //c, enhanced IIe and unenhanced IIe machines — confirmed by the maintainer and by testers. The launcher checks the machine, the ROM, the processor and the memory before starting, says in 40 columns what is missing, and identifies the edition on its title screen.

Choose your disk

Five self-contained images are provided. Choose one from this table; there are no separate category disks to swap when browsing tools.

Image	Contents
A2FILECMD-XL-0.9.2.2mg	Complete 32 MB ProDOS image, 6502, all tools and examples
A2FILECMD-65C02-enhanced-mouse-XL-0.9.2.2mg	Complete XL; 65C02 and enhanced ROM required, mouse support optional
A2FILECMD-DOS3.3-0.9.2.dsk	Standalone DOS 3.3, Apple II+ 48 KB, 40 columns
A2FILECMD-800K-0.9.2.po	Bootable 800 KB ProDOS image, 6502, all tools and BASIC runtimes, no demo corpus
A2FILECMD-140K-0.9.2.dsk	Bootable 5¼-inch ProDOS disk, 6502, essential file operations, editor, text/hex readers, format and verify

The ProDOS editions need 128 KB and 80 columns. mouse denotes optional mouse support; the enhanced edition still works with the keyboard alone. It requires **both** a 65C02 and enhanced ROM. DOS3.3 is a separate program; see its section.

140K and DOS3.3 are 143,360-byte .dsk files in DOS sector order. 800K is an 819,200-byte .po in ProDOS block order. XL is a 32 MB .2mg. Use them directly: changing the extension does not convert an image. With the release images and PDF in one directory, check downloads with:

```
sha256sum -c SHA256SUMS-0.9.2.txt
```

Boot the image, or launch A2FILE.SYSTEM from a ProDOS selector. To install elsewhere, keep A2FILE.SYSTEM beside its complete A2FILE/ directory. Do not mix A2FILE.CODE and native plugins from different builds.

Read the panels

The highlighted panel is active; **TAB** switches sides. Each panel shows names, file types, auxiliary types and sizes. The selected row is highlighted, a star marks a tagged entry, and **L** marks a locked one; both marks sit in the same two columns on every line, a directory's included -- a directory carries a tag like a file. A directory still shows the slash after its name, and <DIR> stands where a file shows its type. The header's star identifies the sort order. Below the panels are free space, selection details, messages and the key bar. **?** opens the help screen.

Returning to a parent with **Escape** or **Return on ..** selects the child you just left, including when it lies beyond the first directory window. The child is located by its current name, so an obsolete index cannot select an unrelated entry. If it has disappeared, the parent opens at its beginning.

The program remembers panel directories, sorting and the active side in A2FILE/A2FILE.CFG when you quit. Saving reserves an exclusive temporary file, verifies every byte after closing it, retains the old file as a backup during replacement, and verifies the installed file before removing that backup. A warning lets you cancel quitting if saving fails. Existing A2FILE.TMP or A2FILE.BAK files are preserved for recovery. This is not power-fail atomicity. On the first XL start, the right panel opens DEM0/; try its text, pictures, music and sample archives. DEM0/CIDERPRESS/ holds real files from CiderPress II's test data, one or more for each format that only a real file shows well (MacPaint, AppleWorks data bases and spreadsheets).

Essential and complete editions

The 140K edition includes navigation, tagging, copying, moving, deleting, renaming, attributes, text editing, text and hex readers, comparison, formatting and verification. It can browse mounted images and extract ProDOS files through IMGFS. It does not include disk imaging, DOS file extraction, archives, media players, document readers or BASIC runtimes. The menu lists available tools. A shortcut to a missing tool reports that it is missing, without asking for an obsolete category disk.

800K and both XL editions include all 82 overlays and both BASIC runtimes. Only XL includes the large demonstration corpus. Use a complete edition for advanced tools. Keep the program and native overlays from the same build; never mix the two CPU builds.

The volume names are /A2FC6502 (140K), /A28006502 (800K), /A2XL6502 and /A2XL65C02. Operations requiring source and destination online still need a second drive or volume. **DISKCOMP S** and **W** → **Copy** in the complete editions retain their single-drive disk exchange modes.

Protect files in /RAM

Copy anything important out of /RAM before displaying DHGR, extracting ShrinkIt, using disk-image operations or physically formatting a Disk II floppy. These operations use auxiliary memory and can rebuild /RAM empty. A warning explicitly says that ALL /RAM files will be lost and asks for consent **before** AUX is used. Declining preserves its contents. The viewer may ask even for a plain HGR file because the same viewer can browse subsequent DHGR pictures; plain HGR itself does not need the reconstruction. The message line reports a reconstruction afterwards. The foreground MB1, PT3 and Electric Duet players preserve /RAM.

Keys

Key	Action
Up / Down	Previous / next entry.
Left / Right, < / >, - / +	Previous / next page; [/] jumps to first / last entry.
TAB / =	Switch panel / show the same directory in the other panel.
RETURN / ESC / /	Open / go up / list volumes. Return asks before running a program.
', then letter or digit	Jump to the next name with that initial.

Key	Action
SPACE / *	Toggle the selection's tag / invert all tags. Directories can be tagged; . never is.
Ctrl-T / Ctrl-N / Ctrl-R	Tag all / untag all / reread both panels.
S / M	Change sorting / tag files absent or different in size/date in the other panel.
C / V	Copy / move tagged entries, otherwise the selection, to the other panel. Includes directories. Move deletes originals after copying.
D	Delete tagged entries or the selection, including directory contents, after confirmation.
R / K	Rename / create a directory. Names: letter first, then letters, digits or periods; maximum 15 characters.
A / L	Change hexadecimal type/auxtype / lock or unlock. Locked files refuse deletion, renaming and writing; the other access bits (read, backup, invisible) are kept.
T / H / I	Read text / hexadecimal / picture. T also lists BAS and AWP files.
E	Edit text; on a directory or . , create a text file.
X	Run a program after confirmation, replacing A2FC.
W / F / I	Disk-image operations / format / plugin menu.
? / 1 ... 0 / Q	Help / key-bar buttons / quit to ProDOS after confirmation.

The mouse

On 65C02, an AppleMouse II supplements the keyboard. Click a row to select it; click the selected row again to open it. Click a path to go up, a column header to change sorting, or a key-bar button to invoke it. Viewers, the editor and prompts use the keyboard.

Copying, moving and editing

Copy or move files

1. Open the destination directory in the other panel.
2. In the source panel, tag entries with **Space**, or leave all entries untagged to act on the cursor alone. **Ctrl-N** clears the tags.
3. Press **C** to copy or **V** to move. Check the destination and answer any replacement question before continuing.
4. Wait for copying and verification, then read the result. On an error, follow Incident recovery before retrying.

Copying preserves name, type and auxiliary type. For an existing target, choose **O** overwrite, **S** skip, **A** overwrite all or **N** overwrite none. Existing destination directories are filled in. Copying into the same or a nested source directory is refused. Files are read back and compared before a move can delete its source. Overwrites protect the old destination as A2FC.BAK; failures attempt to restore it. A leftover backup must be examined before another replacement. Archive and image extraction refuse existing output files instead of silently overwriting them.

Very deep trees can exceed the recursive working space. A2FC checks the remaining stack before directory **I**/**O** and refuses an unsafe traversal with `Directory unreadable or too large/deep`. Copy/move counts the tree before transferring it; directory deletion also scans it before the first removal. Copy or delete smaller subdirectories separately if needed.

Long operations display progress, including each item during recursive deletes. ProDOS copies show separate **Copying...** and **Verifying...** phases; the byte counter restarts for verification and advances during readback. Directory scans and sorting show an activity indicator. Counting files and loading tools announce their phase. A blocking disk call can pause the indicator until it returns. **ESC** interrupts; completed work remains, an incomplete ordinary file copy is removed, and the final message reports what was done. Read the result before removing a disk.

Moving marked files with MOVE

When files are marked, **!** → **Files** → **MOVE** moves the marked files after one confirmation. It reserves and verifies A2MOVE.LST in the destination before moving any source. An existing list is preserved and blocks the operation. Each cross-volume copy is verified completely before its source is removed. The batch stops on the first error or Escape; completed moves remain complete, and pending visible source files are marked again by name. Destination marks are cleared. Marked directories are refused; the existing single-entry MOVE remains available without marks. A retained list is reported for manual review. No auxiliary memory is used.

The text editor

E edits a file; on a directory or **..**, it creates one. The editor holds up to 5,104 bytes, uses CR line endings and strips the high bit on loading. Long lines do not wrap. A star on the status bar means unsaved changes.

Arrows move; **Delete** / **Ctrl-D** erase left / right. **Ctrl-A** / **Ctrl-E** go to line start / end; **Ctrl-P** / **Ctrl-N** changes page; **Ctrl-T** / **Ctrl-B** goes to text start / end. **Return** splits a line; **Tab** inserts four spaces.

ESC opens the menu: **S** save, **X** save and exit, **Q** quit without saving (confirm if changed), **ESC** continue editing.

Saving preserves the file's ProDOS type and auxiliary type. It writes and reads back A2FC.EDIT before installing it, using A2FC.ED.BAK to protect the previous file during renaming. Extra free space is required. If either name already exists, examine/recover it before removing it; A2FC never overwrites a previous recovery file.

Incident recovery

Every ProDOS distribution disk includes **RECOVER**, a text guide at its root. Open it with Return; the help screen (?) also points to it. It identifies COPY, editor, SYNC, conversion and preference recovery files, and explains partial moves and extraction outputs. DOS3.3 has a different writer: its files must not be interpreted using ProDOS temporary names.

1. Stop retrying. Record the error and both full paths. Preserve a disk image or duplicate before modifying anything; work on that duplicate and recover to a separate healthy disk. If the only copy cannot be read reliably, stop. For /RAM, keep power on and refuse operations that reclaim AUX.
2. Keep the target, temporary and backup. Names, dates and matching sizes cannot establish completeness or identify the original target of a generic A2FC.BAK. If the target is unknown, do not guess or restore over a file.
3. On the duplicate, rename a candidate to an unused name such as RECOV.OLD or RECOV.NEW, clear tags, then copy it to a fresh recovery directory. This rename is necessary for A2FC.COPY, a reserved final copy name. Refuse overwrite, and wait for copying **and verification** to succeed.
4. Check the recovered contents; compare every byte with a known good source where available. Only then give the recovered copy its intended name in an empty recovery directory. Retain the image and other candidates until the recovery is confirmed. A temporary may be partial; a backup may be old.
5. For a partial move, compare source and destination file by file. A retained A2MOVE.LST is a work list, not file contents or a safe automatic replay. Preserve remaining sources while recovering missing files.

A2FC does not infer a target name from a generic backup, automatically resume an interrupted transaction, or guarantee recovery after power loss. The bundled guide distinguishes these uncertain states from reported save success with backup cleanup failure.

Recognize temporary and backup names

Names identify possible roles, not guaranteed contents or completion.

Operation	New candidate	Previous version
Copy	A2FC.COPY	A2FC.BAK
Text editor	A2FC.EDIT	A2FC.ED.BAK
SYNC	A2FC.SYNC	A2FC.BAK
Text / image conversion	TXTCONV.TMP / IMGCONV.TMP	A2FC.BAK
Write into a DOS image	A2FC.DOS	A2FC.BAK
Preferences in A2FILE/	A2FILE.TMP	A2FILE.BAK
GOTO preferences in A2FILE/	GOTO.TMP	GOTO.BAK

A2MOVE.LST is a move work list, not a copy of the moved files. Extraction may leave a partial result under its normal output name.

Understand what remains

- **Target and backup present:** installation may have finished but cleanup may have failed. Preserve both until the target has been checked.
- **Backup present, target absent:** the old version may need restoration. Recover it to a separate directory first.
- **Temporary present:** it may be incomplete, or verified but not installed. The name alone cannot distinguish these states.
- **No recovery file:** this does not prove an interrupted write succeeded.

Operation-specific recovery messages

COPY writes A2FC.COPY in the destination directory and verifies its closed contents before replacing the previous destination. Failed cleanup can leave an incomplete temporary; failed installation keeps the verified temporary. If rollback fails, the previous destination remains in A2FC.BAK. Inspect these recovery files before renaming or deleting them. Existing recovery files are never overwritten, and A2FC.COPY cannot be used as the final copy name. No source deletion is authorized after failure, including failed cleanup after cancellation. A power cut can still interrupt physical directory writes.

SYNC stops before processing the next file when installation or recovery cannot finish. Its diagnostic remains visible. Check A2FC.SYNC (the new result) and A2FC.BAK (the previous destination, if moved aside) in the corresponding destination directory. A failed install retains the verified temporary even if the original was restored. A failed cleanup after copying can leave an incomplete temporary. Inspect recovery files before removing or renaming them; retrying does not overwrite them. Source files remain intact.

TXTCONV and IMGCONV report failed cleanup explicitly: the output may remain incomplete, and retrying does not overwrite an existing temporary file. Check TXTCONV.TMP or IMGCONV.TMP in the destination directory (the source directory for in-place text conversion), or the output name when creating a new file. If restoration fails, the original remains in A2FC.BAK and the converted result remains in the temporary file. Inspect these files before renaming or removing them. A retained backup after successful conversion is reported separately. These recovery steps do not provide power-loss atomicity.

See Data safety for failure coverage and physical-write limits.

Reading files and pictures

Text, hexadecimal and AppleWorks

Use **T** for text, **H** for hexadecimal. **Space**, **Return** or **Down** advances a page; **B** or **Up** goes back; **R** restarts TEXT at page one; **R** also restarts BASLIST and AppleWorks; **ESC** returns to the panels. The text reader clips lines beyond 80 columns and remembers up to 96 page starts. Use **MDVIEW** for wrapped text. The hex reader shows addresses, bytes and text. HEX: **G** goes to a seven-digit file offset; **R/E** first/last.

T on a BAS file lists Applesoft line numbers and keywords; on a BA3 (\$09) file, the Apple ///'s Business BASIC. **Return** on a BA3 does the same, and so does **Return** on a name ending in .BA3 whatever type it arrived with -- a Business BASIC program brought from a host keeps its name, not its type, and read as Applesoft the same bytes would be another program. **T** or **Return** on an AWP document reads AppleWorks word-processor text; formatting commands are omitted and tabs expanded.

Return on an AppleWorks data base (\$19) or spreadsheet (\$1B) opens **AWDATA**. A data base shows one record at a time, a row per category, with dates and times as AppleWorks writes them; **TAB** shows categories 23 to 30. A spreadsheet is shown as a sheet: the column letters on the first line, a row of the file a line, each cell in the column the file gives it and at the width AppleWorks saved for that column. Values sit against the right of their column, labels from the left, and a label wider than its column runs into the next ones until a cell writes over it -- so a sentence typed across several columns reads as one sentence. A formula cell shows what the sheet shows: the display string AppleWorks saved, or the result. Only the rows the file holds are drawn, one after another, so an empty stretch does not cost a page; the row number down the left says where you are.

Space, **Down** and **B**, **Up** page through the rows, **<** and **>** (or the horizontal arrows) move one screen of columns, **R** goes back to the first page and the first column. **F** swaps to the cell-by-cell view -- one cell a row, its reference, then its text, its number, or its formula followed by = and the result AppleWorks saved with it --, which is where a formula can be read; **F** again returns to the sheet. Numbers are printed by the Applesoft ROM, as PRINT would; its ninth digit may differ by one from exact rounding. AWDATA reaches the first 2,688 records of a data base and the first 882 rows of a sheet; reports, cell formats and window settings are not shown.

T on an Integer BASIC file (\$FA), including W0Z.BREAKOUT and APPLEVISION, opens INTBASIC.PLG to list its source. **Return/X** executes it through INTBASIC.SYSTEM after confirmation.

MDVIEW also reads extracted Magic Window .MW documents, skipping their 256-byte header and displaying high-bit text. Teach extended files are not supported by the ProDOS 8 reader.

Pictures

Return and **I** use the same picture-format selection on 6502 and 65C02. The viewer is loaded automatically. Explicit packed ProDOS types take precedence; otherwise the first eight bytes identify DGR and HGRR/DHRR without requiring a filename suffix or a ProDOS image type. A failed identification read or close stops opening the file.

Format	Identification	Viewer
Extasie / Chat Mauve	ProDOS type \$F2	EXTASIE
Arlequin / Chat Mauve	ProDOS type \$F8, gs after the two size bytes	ARLEQUIN

Format	Identification	Viewer
MacPaint (576 × 720)	Name ending in .MAC (other names: ! → MACPAINT); the viewer validates the whole file	MACPAINT
Applesoft shape table	Name ending in .SHAPE, or ! → SHAPES on any file	SHAPES
Packed FOT / PackBytes	Type \$08, auxiliary \$4000 or \$4001	PACKFOT
LZ4FH compressed HGR	Type \$08, auxiliary \$8066	LZ4FH
Print Shop monochrome clip art	BIN, auxiliary \$4800/\$5800/\$6800/\$7800, 572 or 576 bytes	PRINTSHOP
MGTK / Apple II Desktop font	Type \$07	FONTVIEW
Hi-res 7 × 8 font (DOS Toolkit, HRCG)	Type \$07, 768 or 1,024 bytes (a BIN of that size: ! → FONTVIEW)	FONTVIEW
Purplesoft GRLOAD pair	Matching .F0T01 and .F0T02, 8 KB each	PURPLE
Packed 816/Paint	Type \$06, auxiliary \$E001 or \$E002	PAINT816
Lo-res page	Type \$06 or \$08, auxiliary \$0400, 1–2,048 bytes	DGRVIEW
DGR with header	DGR signature; the viewer validates the header and payload	DGRVIEW
HGRR / DHRR RLE	Version 1 header, or type \$06/\$08 with .RLE suffix	IMAGE
Raw HGR / DHGR	Type \$06 or \$08, raw page size	IMAGE

Raw HGR accepts 8,192 or 8,184 bytes; raw DHGR accepts 16,384 or 16,376, auxiliary plane first. **Left/Right** browses the raw/RLE album and skips formats handled by other viewers. The panels leave the screen the moment the viewer opens, not when the picture lights up: from there until Escape the screen carries the name being read and then the picture, never the panels. **Escape** returns from every picture viewer. For an unmarked lo-res screen or sprite (BIN/FOT, 1–2,048 bytes), press **I**: DGRVIEW opens and asks for the width when needed. **Return** keeps ordinary small binaries in hex because a sprite has no signature. **H** explicitly opens hex, including for a picture. Unsupported binary formats retain the hex fallback.

FONTVIEW displays the glyphs in code order, sixteen per row. It accepts MGTK fonts with one or two seven-bit columns, up to 128 glyphs and 22 rows per glyph, and hi-res fonts of 96 or 128 glyphs of 7 × 8 dots, a row's high bit shifting it half a dot as on the screen. PRINTSHOP displays 88 × 52 clip art at its 2 × 3 display scale. LZ4FH, PRINTSHOP and FONTVIEW use only main memory and preserve /RAM. They reject truncated data and report read or close errors.

ARLEQUIN shows the pictures of Le Chat Mauve's ARLEQUIN 1.1 graphics interpreter (1985), ProDOS type \$F8: a full-screen picture, or a window such as the demonstration disk's seasons, centred on black. Like EXTASIE it uses the card's mixed mode, colour or black and white chosen byte by byte, and on a machine without the card the ordinary double hi-res colours. The whole file is checked before the auxiliary memory is written; a truncated picture is refused and /RAM is left alone, and after a picture /RAM is rebuilt empty. A \$F8 file without Arlequin's signature opens in hex.

MACPAINT shows MacPaint documents, the Macintosh's 576 × 720 black and white pictures, with or without the 128-byte MacBinary header that files downloaded from Mac archives often carry. The screen holds 560 × 192 of them in double hi-res black and white (eight dots are cut at each edge), and **Up/Down** move the view by 96 lines. An RGB card (Le Chat Mauve, Video-7) is put in its 560-dot black and white mode. The whole file is checked before the auxiliary memory is written; anything that does not unpack into exactly 720 lines is refused, and after a picture /RAM is rebuilt empty.

SHAPES draws an Applesoft shape table (the tables DRAW and XDRAW use) 24 shapes a page, each at scale 1 in the middle of its cell, with the page in the text rows below: **Space/Down** show the next page, **B/Up** the previous one. A shape that does not end within the file stays blank; a file whose shape count is zero, or whose offsets run past its end, is refused. Main memory only: /RAM is left alone.

Left/Right browse the previous/next file handled by the same specialized viewer: Extasie, Arlequin, MacPaint, PACKFOT, 816/Paint, Purplesoft, DGRVIEW, FONTVIEW, LZ4FH and PRINTSHOP. The directory's displayed order is used, including across large-directory windows. At either end the arrow does nothing; Escape returns to the panels. The panels leave the screen as the viewer opens, not when the first picture lights up. While the neighbour loads, the screen keeps what it shows: a hi-res viewer carries the name being read, and a lo-res picture stays on the air in graphics mode until the next one is drawn over it; the panels reappear only when the browsing ends. A viewer that uses AUX asks once before the first AUX write in a browsing session. Left/Right keeps that consent; leaving the viewer clears it.

PURPLE opens either member of a Purplesoft GRLOAD/GRSAVE pair. Keep both files in the same directory, with the same basename: .F0T01 holds the auxiliary plane and the saved mode, .F0T02 holds the main plane. Arrows skip the companion file. The first opening asks for AUX consent; malformed or incomplete pairs are refused. Extended EVE modes (COL280A/B, CP280 and special HGR modes) require compatible EVE hardware/emulation for correct colours; ordinary RGB cards support the usual COL140/BW560 modes. This reader does not yet accept Pascal GLOAD single-file pictures.

Destructive AUX use requires prior consent. Opening a viewer again starts a new session and asks again; declining preserves /RAM. Specialized picture viewers require 800K or XL. Every ProDOS edition holds the internal OPEN.PLG dispatcher; keep it with its matching program build.

The Mockingboard music

Return on an MB1 .MB stream opens the foreground MUSIC overlay on 800K or XL (maximum 4,096 bytes, six voices). **P** pauses/resumes; **Escape** returns to the panels. Playback also returns at the stream's end. The complete file is read, closed and validated before playback. The player uses main RAM and preserves /RAM; an absent Mockingboard is reported.

On Apple //c, a Mockingboard 4c uses the internal connector and answers at \$C400–\$C4FF. The same automatic probe supports it; both MB1 and PT3 use the detected address page. In POM2, enable the card in the //c configuration and use a build containing its Mockingboard 4c support. A plain //c without the card reports its absence.

On the `//c`, waking a 4c can hide the built-in mouse firmware. If that firmware is no longer visible, A2FC disables mouse calls and keeps keyboard navigation available while the card hides that ROM (reset the machine to restore it). This does not affect a slot-based Mockingboard on a IIe, or a `//c` without the 4c.

In all three music players, **Left/Right** select the previous/next tune of the same type in the same directory (MB1 stays with MB1, PT3 with PT3, Electric Duet with Electric Duet), including across large-directory windows. An arrow without a neighbour does nothing. Changing tracks stops the old output and starts the new track unpaused.

Return on a `.PT3` module opens the foreground ProTracker 3 player on 800K or XL. Ordinary modules use three voices on the first AY chip. Standard TurboSound 02TS containers use both AY chips for two independent modules and six voices; playback ends once both modules have finished. **P** pauses/resumes; **Escape** stops and returns to the panels. Playback also stops at the end of the song. A missing card is reported without starting playback. The screen shows the module's title and artist/credit field, then **Player: Vince Weaver - A2FC adapter**. Empty titles use the filename; empty credits show **Not specified**. Fixed-width header fields are bounded and control characters are removed from the display; the module is not modified.

PT3 accepts modules up to **65,535 bytes** using a main-RAM page cache and preserves `/RAM`. The source stays open read-only throughout playback: keep its disk mounted. Slow storage can delay playback when an uncached page is needed. **Esc** also cancels the initial file scan. Frequency tables 0–3 are supported, including their older PT3 3.0–3.3 variants. Multiple deferred special effects within one channel/row are refused. TurboSound accepts the standard two-PT3 container with a 16-byte 02TS footer; the 65,535-byte limit includes both modules and the footer. Other multi-chip container variants are unsupported. At 1 MHz, sparse dual modules can exceed the small cache and slow down even on a hard disk; this implementation does not guarantee 50 Hz for every file. Invalid data, stream pointers, or read/seek/close errors stop loading or playback and return to the panels. Source URLs are listed in `media/pt3/MUSIC/SOURCES.TXT` and at the end of `SAMPLE-MEDIA.md`.

Electric Duet

Return on an Electric Duet song opens the foreground DUET overlay on 800K or XL. A song is recognized by the Apple II DeskTop convention, file type `$D5 (MUS)` with auxtype `$D0E7`, by a name ending in `.ED` whatever its type, or by a BIN name starting with `M.` and a compatible record prefix; the `!` menu runs DUET on any file, which is then validated: complete three-byte records, a terminator before the end of the file, at least one note, and at most 7,168 bytes (the largest known songs are 5.5 KB). The whole file is read, closed and checked before playback, in main RAM; `/RAM` is preserved.

With a Mockingboard (or a Mockingboard 4c on a `//c`), the song plays on the card, one AY tone per voice; without one, it plays on the speaker through Alex Patalenski's improved Electric Duet player, the one Apple II DeskTop uses, which has none of the carrier whistle of the original routine. **1** and **2** switch between the speaker and the card at any time; both outputs follow the same pitch and tempo (the card is timed by its VIA on the speaker player's own unit). **D** cycles the speaker pulse width: 1/8 of the period by default, then 1/4 and Patalenski's original 1/16; wider pulses give more bass on the small speaker, narrower ones let the two voices collide less. **P** pauses/resumes, **Escape** returns to the panels, and playback returns by itself at the end of the song. On the speaker the keyboard is read once per record, so a key acts at the next note. Voice records (the original player's duty-cycle settings) are skipped on both outputs. The demo disk carries `CANON.ED`, Frère Jacques as a two-voice canon generated by `tools/mkdemo.py`.

Disk images and formatting

Disk images

W opens three operations: write the selected image to a formatted target disk, read a disk into a new image in the active directory, or copy one disk to another. Supported containers are `PO`, `DSK/DO` and `2MG`.

For a one-drive copy, choose the same drive for source and destination, then follow the disk prompts. Writing requires **ERASE** in capitals and refuses the running program's disk. Check the named target before confirming. Escape cancels before writing. Disk-image operations can clear `/RAM`.

Disk writes and copies automatically read back every written block. A readback error stops at the first failing block. On one drive, each prompt identifies SOURCE or TARGET copy, the source volume and the slot/drive.

A disk image as a folder

Return on PO, DSK/DO, 2MG or a DiskCopy 4.2 image (.DC, .DC42, .IMAGE, .IMG, the 800K disks of the IIgs and of Mac archives) opens a supported ProDOS or DOS 3.3 image read-only. A DiskCopy image of a Mac disk (HFS) is refused as not ProDOS. Navigate with Return and Escape; Escape at its root leaves the image. **C** extracts tagged files, or the selection, to a real ProDOS directory in the other panel, preserving type and auxiliary type.

ProDOS image extraction supports files up to 128 KB (seedling/sapling). Enter subdirectories and extract their files individually; recursive extraction is not supported. To add a file, use IMGPUT as described below.

Writing into a ProDOS image

On 800K or XL, use IMGPUT to write into a ProDOS image: open it in one panel, put the cursor on a file in the other, and press **C --** or **! -> Disks -> IMGPUT**, which does the same thing. One file, the one under the cursor: tags are not read here, and **V** still refuses, because a move would have to delete the source and IMGPUT deletes nothing. Data is written into free blocks and read back before the allocation bitmap and directory entry are installed. An interrupted operation can leave reserved space; a power failure during a physical metadata write can still damage the image. A file needing more than 128 KB, and a directory with no free slot, are refused rather than half-written.

Reading DOS 3.3 files

A real DOS 3.3 disk appears in / as DOS 3.3, with its slot and drive. Return opens its catalog; **C** extracts files to ProDOS. Applesoft, Integer and binary DOS headers are removed during extraction. Returning to / leaves DOS mode, so the same physical disk can be opened again.

Writing a new file to a DOS 3.3 disk

To **write to a real DOS 3.3 disk from A2FC ProDOS**, open that disk in one panel and select a ProDOS file in the other. Press **C**, then confirm the filename, slot and drive displayed by the confirmation. **DOSWRITE** is on 800K and XL; it can also be launched from **! -> Disks**. It copies the file under the cursor, even if other files are tagged. The ProDOS source must be on another device.

Supported sources are **TEXT, BIN, BAS and INT**, up to 65,535 bytes. BIN gets its DOS load-address/length prefix from the ProDOS auxiliary type; BAS/INT get their length prefix. TEXT bytes are preserved. The destination name is the selected ProDOS name. Existing names, locked or unlocked, are refused. **V** does not delete the source; use **C**. Replacing an existing DOS file is not supported by this command.

Extracting Pascal and CP/M disks

To **read an Apple Pascal (UCSD) disk**, put the cursor on its image (.P0, .DSK, .D0 or .2MG), open a ProDOS directory in the other panel and launch **PASCAL** from **! -> Disks** (800K and XL). It lists nothing and asks nothing: it extracts every file of the volume into that directory, under the one contract the other extractors follow -- a name already taken is skipped and counted, each file is read back against the image before it counts as done, and a failure removes the file it was writing and stops. The image itself is never written to.

A Pascal name becomes a ProDOS one (upper case, anything but a letter, a digit or a period becomes a period, fifteen characters at most). A code file comes out as \$02 PCD, a text file as \$03 PTX, a data file as \$05 PDA, and the rest untyped. A .TEXT file arrives as it lies on the disk, its 1,024-byte header and its page padding included: turning that into plain text is a conversion, not a reading, and PASCAL does not pretend to.

CPM reads an Apple CP/M disk the same way, from the same two panels. A CP/M disk keeps three tracks for itself, then a directory of 64 entries of 32 bytes and files in blocks of 1,024; a file longer than 16 KB has several directory entries, and CPM puts them back in order. CP/M records no exact length, only a count of 128-byte records, so a file arrives rounded up to the record -- a text file ends at the \$1A inside it, as it does under CP/M. Every file comes out untyped; the user number and the read-only and system attributes are not carried over.

CP/M reads the Apple's sectors through an order of its own. The one CPM uses was measured on disks from the Asimov archive, not chosen from the published tables, which disagree; it reads them whole. CPM still does not trust it blindly: it tries its orders and keeps the one whose directory explains itself, so a disk laid out differently is refused ("Not a CP/M volume this can read") rather than read wrongly. Two of the four disks tried are refused that way -- their directories begin elsewhere.

Replacing, deleting and renaming DOS files

To **replace a file on a real DOS 3.3 disk**, arrange the panels as for a copy -- the ProDOS file selected, the DOS disk opposite -- and launch **DOSREPL** from ! -> **Disks** (800K and XL). It writes the new copy into sectors of its own, reads it back, and only then switches the catalog entry to it in a single sector write; the old file stays whole and readable until that instant. The disk therefore has to hold both copies at once, and says so when it cannot. An interruption before the switch leaves the original untouched; after it, the old sectors may stay marked in use, and the message says so rather than calling the work done.

To **delete or rename a file on a real DOS 3.3 disk**, open that disk in the active panel, put the cursor on the file and launch **DOS33W** from ! -> **Disks** (800K and XL); **D** deletes, **R** renames, each after a question naming the file and the drive. The file is identified by the track and sector of its first track/sector list, not by the name on screen, which is a ProDOS-shaped copy of the DOS name. A locked file, a write-protected disk, a drive that is not a standard Disk II, and any volume whose catalog and track/sector lists do not add up are refused before the first write -- and the same audit runs again after the question is answered, because the drive door is open while it waits. Deleting writes the catalog entry first and the bitmap after, so an interruption between them can leave sectors reserved. The message reports that state. Preserve the disk for inspection; **FIXIT** checks ProDOS volumes, not DOS 3.3 allocation. A physical sector write is not atomic. A new name is read by the usual prompt, so it holds letters, digits and periods: DOS 3.3 itself allows more, including the comma that breaks its own CATALOG listing.

The target must be a standard 35-track, 16-sector DOS 3.3 disk in a Disk II in any slot 1-7, with the standard Disk II ROM signature. The command checks physical write protection, the VTOC, catalog, all live sector lists and allocation conflicts before writing. It reserves sectors first, verifies each written block and compares the source again before publishing the catalog entry. No auxiliary RAM disk storage is used. After an interrupted or failed write, the source remains intact, but some space may stay reserved. Check the DOS disk before further work. Physical VTOC/catalog writes are not atomic: a power cut or damaged sector can still corrupt shared metadata.

Writing into a DOS 3.3 image

To **write into a DOS 3.3 image**, open its catalog with Return in one panel, select the ProDOS source in the other, and press **C**. Confirm the source name and complete target image path. Supported containers are **.DSK**, **.DO** and DOS-order **.2MG**, holding a standard 35-track, 16-sector disk. A locked 2MG container or protected ProDOS image file is refused.

The 800K/XL helpers create **A2FC.DOS** beside the image, copy and compare every byte, then perform the same DOS allocation audit and verified copy on that temporary. The closed result is reread before installation, and the original image is checked again for changes using its full length and CRC-32. The original is renamed to **A2FC.BAK** before the result takes its place; an installation failure attempts to restore it. Existing temporary/backup names are never overwritten. If recovery files are reported, keep them until checked. The containing ProDOS volume needs enough free space for a complete second image. No source file is deleted and no AUX RAM disk storage is used. This replacement preserves recovery files on reported errors, but is not atomic across a power cut.

DOS extraction and old Electric Duet files

C extracts DOS BIN/BAS/INT to the exact length declared by the DOS header. BIN retains its original load address as the ProDOS auxtype; BAS uses \$0801. Sector padding and the DOS header are excluded. TXT retains its sector data. The DOSGET overlay requires 800K or XL. Reading a DOS 3.3 catalog as a panel uses the internal CATALOG overlay, on every ProDOS edition, which also walks the directories of a disk image opened as a folder. An existing destination is refused. An I/O failure stops extraction; a failed cleanup names the newly created file that remains. Completed files survive a later source-close error, which is reported. Escape cancels between sectors.

Return also tries DUET for BIN names beginning M. with a compatible record prefix; the player validates the loaded records before producing sound. This does not change attributes. For old BIN/\$0000 extractions, mark the songs with Space, select ! → **Files** → **FIXTYPES**, then confirm each proposal to \$D5/\$D0E7. FIXTYPES reads the full candidate, requires a complete terminator and at least four audible notes, validates control records, and accepts up to 255 legacy padding bytes. It preserves the original name and all content, including padding. N cancels that file before any metadata write.

IDENT now covers DUET, PT3, MGTK fonts, Purplesoft pairs, DGR/lo-res, Print Shop and the specialized packed picture formats as well as its existing families. These are format hints or structural matches, not proof for every possible file: several raw formats have no unique signature. Ambiguous or damaged DUET candidates are reported as unrecognized; no attributes are changed by IDENT. Read, seek and close errors are reported as errors.

NIBCOPY: physical 5¼-inch copies

Open ! → Disks → NIBCOPY from 800K or XL. Select the Disk II slot, source drive and target drive. Selecting the same drive enables exchanges. Use normal 1 MHz speed and disable accelerators for this timing-sensitive transport. Cover the source disk's write notch **before** starting and keep it covered: NIBCOPY refuses an unprotected source, and refuses to write a protected target. This also prevents writing the source accidentally during a single-drive exchange.

The loader first warns that **all /RAM files will be lost** and requests permission before NIBCOPY uses auxiliary memory. Save those files elsewhere before accepting. NIBCOPY keeps its code and buffers in RAM; the program disk can be removed while copying. Remove it before inserting the source and target. Confirm destruction of **all target files, including locked files** on the displayed slot/drive before the first write. In single-drive mode the same confirmation is required after every target exchange. Return accepts a source insertion; it does not confirm target destruction. On exit A2FC may ask for its program or panel disks again.

Each source track is read twice. Both complete sets of 16 address/data fields, their checksums and circular sector order must agree before writing. The target is then reread and all encoded fields compared. Escape stops between tracks or cancels an insertion/confirmation. The final report gives the number of verified tracks out of 35; any read, format or verification error stops the copy. An incomplete target must not be treated as a valid backup.

This initial transport supports standard 16-sector Disk II framing only. It preserves encoded payloads and sector order but regenerates synchronization gaps; it is **not** a flux copier or a preservation tool for copy protections, 13-sector, half-track, weak-bit or other nonstandard formats. Unsupported or unstable source tracks are refused before writing that track. It does not preflight all 35 tracks before the first write. A write failure or power cut can leave a partly overwritten target; no rollback or power-failure atomicity is promised. Hardware qualification on real drives and accelerated machines remains necessary; native automated tests cover both IIe CPU variants in POM2.

Formatting a disk

Press **F** or choose **FORMAT** in !. Select the drive, enter a volume name, then type **ERASE** and press Return. The list shows slot, drive, volume and capacity. The program's own volume is marked **IN USE** and cannot be formatted. Escape or a different confirmation word cancels.

Disk II formatting clears /RAM; move its files elsewhere first. It is refused if A2FC itself runs from /RAM. Formatting another block device does not require that extra RAM reset. The result reports success or an error; Escape returns directly to the panels.

Archives and programs

Unpacking archives

Set the destination in the other panel, select the archive, then use !:

Tool	Supported archives
UNSHRINK	ShrinkIt .SHK: stored data, LZW/1 and LZW/2. Files retain ProDOS type and auxiliary type, then their archived lock and modification date once read back; names are adapted to ProDOS. Disk-image members become PO files. Resource forks and comments are skipped, and so is a file whose type or auxiliary type ProDOS cannot hold, or a disk image not made of 512-byte blocks. Each file is decoded a second time and compared with the file read back; a mismatch removes it.
BINARY2	Binary II .BNY/.BQY: extracts members with their names and attributes. Directory entries are skipped. Compressed members may need a second extraction with UNSHRINK.
SCIIBIN	BinSCII text (.BSC, .BSQ), the Usenet encoding of Apple II files: decodes the file it carries into the other panel, with its name and ProDOS type. A file posted in several parts: select the first one; the files after it in the directory are read while the file is incomplete. Every chunk's header and data CRCs are checked, the parts must follow each other in order, and the result is read back against the CRCs. Failed extraction attempts to remove the new output; a cleanup failure is reported. Return opens .BSC and .BSQ files.
UNSQ	SQueezed files (.QQ, as BLU makes them; a .BQY archive's members come out of BINARY2 as such files) and AppleLink ACU archives (.ACU): extracts into the other panel under the stored names (a .QQ keeps its own ProDOS type; ACU records carry theirs), directories skipped and paths flattened. A taken name is skipped and counted; a damaged stream or a failed write removes that file and stops. A .QQ is checked against its checksum; ACU's data CRCs are not trustworthy, so its lengths are checked instead. Every file is read back. Return opens .QQ and .ACU files; ! opens any other.
UNWRAP	AppleSingle (\$E0/\$0001 or .AS, versions 1 and 2) and MacBinary I/II/III: extracts the data fork into the other panel under the wrapped file's name, made a ProDOS name (or the wrapper's, less its suffix), with its ProDOS type -- from the ProDOS information, or converted from the Mac type and creator as AppleShare does. The file is created only if the name is free, read back and compared, and removed on failure where cleanup succeeds. The resource fork is left out, and the note says so. Return opens AppleSingle files; use ! for MacBinary.

ShrinkIt extraction clears /RAM and refuses it as a destination. Keep archives and recovered files on another volume. A warning requires confirmation before AUX is touched. **Escape** cancels extraction between blocks. Existing files are never replaced. On error or cancellation, only the current new file is cleaned up; if cleanup fails, the message names the retained file. A retry refuses that file without truncating it. Completed earlier members remain available. Truncated skipped threads, stream errors and failed closes are reported instead of a successful extraction. An unsupported member ("Unsupported file skipped.") remains visible in the final message even if other members were extracted. Every CRC the archive carries is checked: the master header and each record header before anything of that record is written ("Header CRC error: archive not trusted."), the LZW/1 stream and, from record version 3, each thread's data. A file whose attributes cannot be set stays, verified, and the message says so ("attributes not set").

Running BASIC and machine-language programs

Return runs BAS, INT or SYS after confirmation; **X** also runs BIN. The launched program replaces A2FC; it does not automatically return. A BIN loads at its auxiliary address, which must be between \$0800 and \$BAFF.

Applesoft (\$FC) requires BASIC.SYSTEM; Integer BASIC (\$FA) requires INTBASIC.SYSTEM v0.9. Both are supplied on 800K and XL; the runtime is chosen automatically. **T** lists either BASIC source without executing it. A2FC searches the program's volume, its own volume and a tool disk if present. Use 800K or XL for the supplied BASIC runtimes. An unreadable or incompatible runtime is refused. With one floppy drive, keep the BASIC program on another online volume so it remains readable after the runtime loads.

Launch checks use the file's actual size and keep the load below \$BB00, where the loader's ProDOS I/O buffer starts. If the program disk is full and preferences cannot be saved, A2FC asks whether to run anyway; it does not silently discard that failure. Integer BASIC returns to the ProDOS selector at program end. Its upstream runtime supports a subset of DOS commands; see INTBASIC.SYSTEM usage and provenance.

The confirmation before a launch spells out the way back, for example Run HELL0? Back: -/A2FC6502/A2FILE.SYSTEM: from the Applesoft] prompt, reinsert the program disk if needed and enter that command. BASIC.SYSTEM keeps its prefix on the launched program's directory, so the bare -A2FILE.SYSTEM only works when that directory is A2FC's own. BYE also leaves to the ProDOS selector, where A2FILE.SYSTEM can be picked. BAS paths over 46 characters use a fallback launch method; use a shorter path if launch fails.

More tools in the ! menu

Select the item first, press **!**, choose a category and press **Return**, then choose its tool and press **Return**. Categories are Files, Images, Music, Disks, Programming, System, Archives and Other (third-party tools). **Escape** returns to the category list, then to the panels. Tools are sorted by name within each category. **Up/Down** moves a line, **Left/Right** moves six lines (stopping at the list ends), and a letter jumps to a matching initial. Questions on the penultimate line appear in inverse video while awaiting a response, including typed names, confirmations, conversion choices and disk swaps. Ordinary information and results remain in normal video. The following tools supplement the main keys and readers.

File and volume tools

Tool	Operation
COMPARE	Compare the selected file byte by byte with the same name in the other panel.
TXTCONV	C = CR, L = LF, D = CRLF, H = clear high bit, S = set it, T = expand tabs, A = transliterate UTF-8 accents; incomplete sequences become ? without consuming following text. Write in place or to the other panel. In-place conversion preserves the source on incomplete reads or changed size and refuses an existing TXTCONV.TMP.
DATE	S sets date/time from DDMMYYYYHHMM (1940–2039); impossible dates are rejected. F stamps modification dates on tagged files or the selection. Creation dates stay unchanged; a hardware clock may replace the entered time.
VERIFY	Read tagged files (skip directories), the selection, or every block of a volume. Report processed files and errors; ESC cancels. No writes.
TAGPAT	Name patterns: = any string, ? one character. Add comma-separated filters: T04 TXT, >2000 or <2000 bytes, D modified today. T tags, U untags, X replaces tags.
WIPE	F checks live directory/file references against the bitmap before zeroing free blocks. Unreadable, inconsistent or unsupported allocation is refused. W zeroes the whole volume after ERASE; the running program's volume is refused.

Complete-edition tools

VOLINFO, FIXIT, REPAIR and VOLNAME are included on 800K and XL. Of the preceding table, 140K includes COMPARE and VERIFY; TXTCONV, DATE, TAGPAT and WIPE require 800K or XL. Menu categories describe tasks and do not require changing disks just to browse.

Tool	Operation
VOLINFO	Audit allocation and fragmentation. M = bitmap (. free, # used), F = selected file blocks, E = export to the other panel. N/P pages; ESC returns. No repairs.

Tool	Operation
FIXIT	Check a ProDOS volume and name each fault: header, directory chains and parents, entry names, access bits, key and index pointers, file and directory counters, cross-linked, lost and wrongly marked blocks. One line per check with its count and first block; 18 lines a page, a key continues. R scans again after a disk change; ESC/Return leaves. On a volume above 4,096 blocks it first asks for Q (quick: directories only) or F (full), then whether /RAM may be lost. FIXIT only reads: it writes nothing and repairs nothing. REPAIR is the tool that writes.
REPAIR	Repair a ProDOS volume. It walks the volume itself, shows a plan -- one line per check with the number of corrections, plus what it refuses and why -- and writes nothing until F is pressed and the word FIX typed in full. It repairs eleven faults: in the bitmap, blocks a file uses but the bitmap calls free, reserved blocks marked free, bits set past the end of the volume and blocks nobody claims; in the directory tree, a header's file count, a file's or a subdirectory's blocks used, a subdirectory's eof, an entry's pointer back to its own directory, the three parent fields of a subdirectory header and a directory block's back-pointer. Every block written is read back and compared; a block that cannot be verified has its original rewritten and verified. The volume is then walked again and the verdict says what that second pass found.
VOLNAME	Rename a ProDOS volume and update the affected panel/program paths. A name another online volume already has is refused ("Name in use."); ProDOS itself would accept it, and two volumes would then answer to one path.
SEARCH	Find text in the active directory and tag matching files, ignoring case. ESC cancels a long scan and keeps tags already found.
FIXTYPES	Review and confirm type/auxtype repairs on tagged files or the selection. Recognizes validated DUET content with a name/type hint and explicit suffixes; optional suffix removal. DUET names, image suffixes and .SYSTEM stay.
GOTO	P opens a typed /VOLUME/DIRECTORY path (63 characters max; Delete/Left edits, ESC cancels). Nine favourites: A adds, D then a digit removes, M then two digits reorders, 1–9 jumps. Saved in A2FILE/GOTO.CFG.
FIND	Search the volume by name pattern; start with " to search contents, ignoring case. TAB sets type (T, two hex digits) and modification dates (D, inclusive YYYYMMDD, 1940–2039); A clears filters. Undated files are excluded by date filters. N shows the next 20 results; Return jumps there. V on a text result shows occurrence offsets (hex) and excerpts; N/Space continues, ESC returns.
BLKVIEW	Read device or image blocks: H hex/ASCII, D directory, I index, N/P block, Space page, G four-digit hex block, F find four bytes (8 hex digits), A find next, X extract blocks, ESC back. Source stays unchanged.
ARLEQUIN	View ARLEQUIN/Chat Mauve ProDOS \$F8 pictures, full screen or windows. Return and I select it when the file carries Arlequin's signature.
EXTASIE	View Extasie/Chat Mauve ProDOS \$F2 images. The original count/repeat stream is decoded into the HGR page; ESC returns to the panels. Return and I select EXTASIE automatically on both processors.
DISASM	Read BIN/SYS as assembly: N/Space next, P previous (last 64 pages), C 6502/65C02, G seven-digit file offset, L four-digit CPU load address, R start, E export, ESC back. BIN uses its auxtype; SYS starts at \$2000.
CRC	Calculate CRC-32 for the selection or tagged files. Results appear in pages of 20; a key continues, ESC at a page boundary stops the batch.
IDENT	Identify supported music, picture, font, archive, program and disk families from headers, attributes or names; DUET candidates are read completely. Text statistics cover the first 512 bytes.
MDVIEW	Wrapped Markdown/text; no forward limit. Up: last 64 pages. R: restart.
RENAME	Batch prefix, suffix, extension replacement/removal or numbering. For example E then BAK sets .BAK. Conflicts are skipped.

Tool	Operation
IMGCONV	Convert PO/HDV, DSK/DO, 2MG and DiskCopy 4.2 (C; \$E0/\$8005, 400K/800K/720K/1440K only) into the other panel, preserving disk blocks. A DiskCopy source is checked against its own checksum before anything is written; its tag bytes are not converted. Unsupported 2MG formats, block counts exceeding 16 bits, and data ranges inside the header or beyond the source size are refused before destination access. Read or seek failures abort conversion and attempt to remove incomplete output; failed cleanup is reported.
BOOTBLK	Copy ProDOS boot blocks from the boot volume to another volume after confirmation. Saves both originals in main memory, verifies writes and restores both blocks on error. An incomplete restoration is reported explicitly; the backup does not survive a power cut.
UNDELETE	Browse deleted ProDOS entries. N skips; R recovers a validated candidate to another online volume. Existing names are refused.
NIBCOPY	Physical Disk II copy, one or two drives. Copies 35 standard 16-sector tracks, retaining encoded fields and sector order, regenerating sync gaps, and verifying each track. Requires a write-protected source and prior AUX/target confirmations.
DISKCMP	V compares online ProDOS volumes; I compares images; S compares two Disk II disks on one drive. Reports differing blocks and the first mismatch.
MKIMAGE	Create an empty ProDOS PO or 2MG: 140 KB, 800 KB, 2/4/8 MB or 32,767 blocks. New images are data volumes, without a boot program.
RESCUE	F recovers a file; V recovers a ProDOS volume. Uses up to 30 attempts per block, zero-fills unreadable chunks and writes a LOG. Destination must be another online volume.
SYNC	Recursively copy missing or newer files to the other panel after confirming direction. Destination-only files remain; copies are read back before replacement.
MOVE	Move marked entries, or the selected entry without marks. Within a volume, move without copying data blocks, directories included; locked sources are refused. Every path component must still be a directory. A full subdirectory grows if space is available; damaged parent references are refused before writing. Across volumes, a file is copied and verified before its source is deleted; existing destination names are refused, and a size mismatch preserves the source and removes the incomplete copy (if that removal fails, the message says the partial copy stays). A directory bound for another volume, marked or under the cursor, is walked like V does: counted, copied and read back file by file, and its source deleted only once every file has arrived; a copy that stops, or a skipped file, keeps the whole source and stops the batch. Available on every ProDOS edition.
TREE	Show file sizes and cumulative directory totals. Space advances a page; ESC exits.

Tool details and limits

SYNC and TREE

SYNC skips equal-date or newer destination files. A source with an unknown date does not replace an existing file. It uses A2FC.SYNC temporarily and A2FC.BAK for rollback, preserving pre-existing files with those names. Verification requires matching contents and exact length; a size mismatch preserves the source and existing destination and removes the temporary copy. If replacement fails, the original is restored where possible; keep any remaining A2FC.BAK. Overlapping directory trees are refused. SYNC and TREE support paths shorter than 64 bytes and up to 16 directory levels; read errors and unsupported resource forks produce an error/incomplete result.

DISKCMP

DISKCMP S buffers two blocks per exchange, so a full comparison needs many swaps. Each prompt names the expected disk and slot/drive; **1/2** changes the drive, **Return** retries and **Escape** cancels. A read error or cancellation never produces an "identical" verdict. Image comparison supports PO/HDV, DSK/DO and ProDOS-order 2MG.

REPAIR

REPAIR refuses what it cannot settle. Cross-linked blocks stop the plan whole -- not one bitmap page, not one counter: a block two things claim may be the block a counter repair rewrites while a file holds it as data, and nothing can tell which claimant owns it. Copy both files to another volume first, as the message says. A file entry the walk had to abandon (a key or an index pointer out of range, an impossible storage type) stops the repairs of the bitmap that would free a block: the blocks nobody claims may be that file's tail, which **RESCUE** and **UNDELETE** can still read. That one does not stop the other repairs -- directory corrections and the bitmap pages that only mark a block used are still written. A refused header, a read error, a directory loop or a pass **Escape** cut short refuse the plan whole. So does the volume the program itself is running from, before a single block is read: ProDOS 8 keeps a bitmap block of its own in memory and would write it back over the repair. Between the plan and the first write the header of block 2 is read again and compared, all thirty-nine bytes of it, so a floppy swapped while the question was on the screen receives nothing. The forward chain of a directory is never rebuilt, only the back-pointer. **REPAIR** names only what it can repair or refuse: file names, access bits, oversized eofs, the volume name and the shape of the volume directory are **FIXIT**'s business.

Each correction is one verified write, so a block carrying several of them is written once per correction; each write keeps the block as it was in main memory, reads it back and compares all 512 bytes, and puts the original back if anything differs. Those originals live in RAM only: they do not survive a power cut, and ProDOS offers no transaction over several blocks, so a plan interrupted leaves part of it applied. The verdict says repaired only when the second pass comes back with nothing at all; otherwise it counts what it still sees. Every volume is walked three times, once for the plan, once to write and once to check, and **REPAIR** shows no progress line while it walks: about seven minutes for a full 32 MB hard disk on a 1 MHz machine.

FIXIT

FIXIT examines a real ProDOS volume only; an image or a DOS 3.3 disk opened as a directory is refused, and the volume must be on line. A read error, a directory loop, more than 16 levels or a refused header stops the pass: it then says so and never reports lost blocks, which may belong to the part of the tree it could not reach. Only the first sixteen findings keep a block number; beyond that a check shows its count alone. Nothing is written, on the checked volume or anywhere else, and no report is exported: **REPAIR** is the tool that writes.

Large-volume scans and /RAM

Volumes above 4,096 blocks. **FIXIT** and **REPAIR** keep one bit per block of the whole volume, 8 KB, in auxiliary memory, where ProDOS keeps the /RAM disk. Before touching it they ask "ALL /RAM files will be LOST. Continue?" once per run; **N** leaves with "Scan cancelled" and nothing read beyond the volume header. On the way out /RAM is rebuilt empty. A volume in slot 3, drive 2 -- where /RAM lives, and where a larger RAM disk in auxiliary memory replaces it -- is never checked this way. **FIXIT** asks first how deep to look, at every pass (so **R** can follow a quick check with a full one): **Q** reads the directories and nothing else -- names, access bits, counters, chains, parents, key pointers, eofs -- in seconds, and its title ends with -- **QUICK**; it does not check index blocks, extended files, block counts, cross-links or the bitmap, and a clean result says "Directories consistent (quick check)". **F** checks everything, about two minutes and a quarter for a full 32 MB hard disk on a 1 MHz machine. **REPAIR** always does the full walk: without the file blocks it could not tell whether a directory block it rewrites is also held by a file.

VOLINFO and MKIMAGE

VOLINFO supports ProDOS files, both forks and directories up to 16 levels. Large volumes take longer; incomplete counts are unconfirmed. File lists show data, index, master and extended blocks, omitting sparse holes. Exports include the selected file and describe the scan before report creation; existing names are refused. Only complete exports end with `END REPORT`; partial files remain. **MKIMAGE** refuses existing names and removes cancelled/incomplete new images; 32,767 blocks is the maximum that fits in a single ProDOS image file.

DISASM

DISASM shows file offsets, 16-bit CPU addresses, bytes and instructions. `C` changes decoding at the current offset and resets page history; `65C02` includes Rockwell/WDC extensions. Unknown/truncated instructions appear as `.BYTE`. `E` exports from the current offset to EOF as a new `TXT` file in the other panel, using the displayed CPU and load address. Existing names are refused. `ESC` cancels; errors or cancellation keep partial output. Exporting preserves the current page.

BLKVIEW

BLKVIEW F searches forward from the current block, including matches across block boundaries; `A` continues without wrapping. `X` extracts from the current block: enter a four-digit hex count (maximum `7FFF`) and a new filename in the other panel. Device sources require another destination volume. Errors retain partial output.

UNDELETE and RESCUE

UNDELETE never changes the source directory, indexes or bitmap. It checks retained pointers, block counts, free blocks and index halves swapped by ProDOS `DESTROY`. An interrupted `DESTROY` can leave a deleted entry with damaged allocation information: the deleted marker alone is insufficient. Reused, inconsistent or ambiguous candidates are refused. Standard seedling, sapling and tree files, including sparse files, are supported; deleted directories and resource forks are not. Inspect recovered data before relying on it.

RESCUE writes `BASE.REC` for a file. Whole-disk recovery writes raw ProDOS parts `BASE.P01`, `BASE.P02`, etc., up to 16,000 blocks each. Concatenate them in numeric order for a `PO` image; a single part can simply receive `.P0`. `BASE.LOG` records zero-filled chunks and completion or interruption. Partial output stays after cancellation or a write failure.

VDrive: two volumes over the serial line

A detected Super Serial Card or `//c` serial port can expose two remote ProDOS volumes at **115,200 bps**. Slot 2 is tried first. The status line identifies the serial card and assigned drive slots; the remote volumes then work like local ones for browsing and copying.

Use ADTPro's virtual-drive server, `veserver.py` or `sur1-server`; the host supplies date/time during reads. Disconnection produces an I/O error. The driver is removed on quit or program launch.

Limits and troubleshooting

Symptom or limit	What to do
CPU or memory refusal at startup	Use the matching CPU build, with 128 KB and 80-column support.
Missing or stale plugin	Use 800K or XL for advanced tools. Keep <code>A2FILE.CODE</code> and its native plugins from the same build together.

Symptom or limit	What to do
GOTO.TMP or GOTO.BAK remains	A failed install keeps the verified GOTO.TMP; failed cleanup before verification can leave an incomplete temporary. If GOTO.CFG is missing and GOTO.BAK is present, inspect and restore the backup. Inspect all recovery files before renaming or deleting them; retrying does not overwrite them.
Disk changed but old contents remain	Press Ctrl-R to reread both panels.
Run failed: file not found	Check the selected program, its path and BASIC.SYSTEM for BAS files.
Image cannot be opened	Check its actual format; renaming PO to DSK does not convert it. Some file/image features are unsupported.
Directory exceeds 139 entries	A2FC uses windows in disk order, without sorting. Cross a window edge to continue.
Very deep directory copy refused	Simplify the tree or copy smaller subdirectories; recursive operations have bounded working space.
Incomplete scan or recovery	Read the reported limits or errors; inspect recovered data and keep originals/backups.

make disk builds both CPU families; make test runs host checks. Development: README, sdk/README.md, bench/README.md. Remaining work: TODO.md.

A2 File Cmd DOS3.3

A standalone edition for an **Apple II+ with 48 KB and an NMOS 6502**: two panels in 40 columns and DOS 3.3 copying between two Disk II drives. No ProDOS, no 80-column card, no auxiliary memory, no language card. It is a separate program on its own disk, A2FILECMD-DOS3.3-0.9.2.dsk; nothing in the sections above applies to it, and it carries no overlay or plugin. Written entirely in 6502 assembly. The developer guide, measurements and build notes are in MINI-DOS33.md.

Booting

The disk boots through an Applesoft HELLO that shows A2FILECMD, MINI DOS 3.3 and V0.9.2, then BRUN A2FC. From DOS 3.3, use BRUN A2FC. Both panels open on the boot disk; each remembers its drive, selection and scroll position. Each panel shows 19 rows and a catalog of up to 105 files; ? lists every control.

Keys

Key	Action
Tab (Ctrl-I), or 1	Switch panels
Up / down (Ctrl-K / Ctrl-J), or I / K	Previous / next file
Left / right, or - / +, or < / >	Previous / next page (19 files)
[/]	First / last file
Return, or 2	Open by content: text, hi-res picture, hexadecimal, or BRUN NAME? for a program (see below)
T / H	Text / hexadecimal preview; also available inside the preview
G	Hi-res viewer: first 8 KB of the selected file
B	BRUN the selected binary after Y: A2FC leaves, then DOS runs it from the panel's drive

Key	Action
Space	Tag or untag the selected file
Ctrl-T / Ctrl-N / *	Tag all / none / invert on the active panel
N	New text file: name, then the editor
E	Edit a text file in RAM, then save under a new exclusive name
D	Delete tagged files, or the cursor if nothing is tagged
L	Lock or unlock: the cursor toggles; tagged files unlock if any is locked, or lock if all are unlocked
R	Rename the cursor file (exclusive new name; locked files refused)
C, or 3	Copy tagged files, or the cursor if nothing is tagged
/, or 4	Switch the active panel's drive and reread it
Ctrl-R, or 5	Reread both panels, preserving selection by name
=	Show the same disk in the other panel
Escape in preview, help or editor	Return to the panels
?, or 6	Keyboard help
Q, or 7	Ask to return to DOS 3.3

Return reads the file's first data sector, writes nothing, and picks the view from what the file holds:

- **T**: the text viewer, or hexadecimal when the bytes are not text.
- **B** whose DOS header (load address and length) matches the file's size: an 8 KB load at \$2000 or \$4000 opens in the hi-res viewer; an empty file, a load below \$0800 or one reaching DOS's buffers at \$9600 opens in hexadecimal; text opens in the text viewer, after the 4-byte header; anything else is a program and asks BRUN NAME?.
- **B** without a matching header: a raw hi-res page at 32–34 sectors, otherwise hexadecimal.
- **A, I, S, R**: hexadecimal.

Bytes read as text when at most one in 16 is neither printable nor RETURN. T, H, G and B still force the text, hexadecimal or hi-res view, or BRUN.

Questions appear in inverse video on the footer: **Y confirms, N or Escape cancels**, every other key is ignored. Copy, delete, lock, rename and create return to the panels on their own; the result takes the file name line until the next key. The last row shows the main keys in inverse, each with its action attached, without repeating the key's letter: TAB PAN, C OPY, D EL, B RUN, / DRV, ? HELP, Q UIT; questions show Y ES, N O, ESC ANCEL. Only panels showing the written disk are reread; two views of that disk share one catalog. Ctrl-R still rereads both. Escape at the browser does nothing: DOS 3.3 has no parent directory.

Copying, and every other disk write

1. Give the two panels **two different drives** (Tab and /).
2. Tag the files (Space, Ctrl-T / Ctrl-N, *), or leave the cursor alone, and press C. The footer asks COPY name? or COPY N MARKED?.
3. Each file is created exclusively: an existing name is skipped and the batch goes on. A [-----] bar fills while the file's sectors are written and read back. An uncertain write stops the batch.

A copy preserves the name byte for byte, the type, the lock flag and every byte of the data sectors. The source is never written or deleted. Required sectors are reserved in the destination VTOC before any data, each written sector is read back, and the catalog entry is published last, only if its slot is still free.

N and E create text files with the same engine: one new name, never an overwrite. E loads at most 32 data sectors (8 KB) and refuses a larger file rather than save it truncated. L writes only the catalog type byte; R only the catalog name bytes, to a name that does not exist yet; a locked file must be unlocked first. D marks the catalog entry deleted first, as DOS does, then frees the sectors; locked files are skipped.

Every write goes back to the **catalog slot the panel read** and holds the disk to it: the T/S pointer, type, sector count and name must still match, or the operation is refused as **DISK CHANGED** before any write. A write-protected disk is refused as such. A looping catalog chain is refused rather than followed.

Limits

Standard 35-track, 16-sector DOS 3.3 disks, with multiple T/S lists per file. Sparse or inconsistent chains, and data on track 0 or the catalog track, are refused; 13-sector, 40-track and protected formats are not supported. Two drives on the boot controller are required: no single-drive copy by swapping disks. After Y, the keyboard cannot interrupt. A disk swapped at the prompt is refused, but a byte-identical twin is not caught. A DOS sector write is not atomic: a power loss while the VTOC or a shared catalog sector is written can damage metadata; readback detects errors, it cannot repair them. An uncertain error leaves reservations in place and blocks further writes for that run: have the disk checked before reuse. Previews show the first 256 stored bytes only. After changing a disk, press Ctrl-R to reread both panels.

Progress and timing

The copy engine reads sectors in batches, then writes and verifies them on the destination. Watch the progress bar and wait for the final result before removing a disk. Measurements and implementation details belong to the DOS3.3 developer guide; emulator timings are not a promise for every physical drive.

Credits, inspirations and reused code

A2FileCmd is original GPL v3 software by **Arnaud Verhille**. The links below identify the projects that shaped its interface, supplied technical references, or contributed code patterns. They are listed so that the provenance of every borrowed or adapted fragment is easy to check.

cc65 — Oliver Schmidt Toolchain and Apple II startup code. `src/crt0.s` and `src/crt0_loader.s` retain the V2.19 provenance notice; local loader changes are marked there. URL: [cc65 repository](#) · [Apple II startup source](#)

Colin Leroy — a2tools / Ammonoid The serial virtual-drive glue in `src/vsdrive.s` follows the documented a2tools approach; the source comment names the corresponding file and author. URL: [a2tools repository](#) · [Ammonoid releases](#)

Jerry Hewett and Gary Desrochers — Hyper-FORMAT The Disk II ProDOS formatter descends from their public-domain routines, carried by ADTPro and credited in `src/format.c` and `src/format_diskii.s`. URL: [ADTPro source tree](#)

ProDOS 8 documentation File-system structures, MLI calls, allocation blocks and path rules follow the published Apple II technical references. URL: [ProDOS 8 technical information](#)

A2Command The Apple II two-panel file-manager model and command-oriented disk workflow are explicit design inspirations. URL: [A2Command archive](#)

Norton Commander The two-panel layout, selection marks and bottom key bar are interface inspirations. URL: [Norton Commander archive](#)

ShrinkIt / NuFX and Binary II UNSHINK implements documented LZW/1 and LZW/2 formats; BINARY2 follows the public member-header format. No archive executable code is bundled. URLs: [NuFX notes](#) · [NuLib format library](#)

Paul Lutus — Electric Duet The Electric Duet song format (three-byte records, two voices) played by DUET. Its original player routine is under the GPL; the format is documented by its author. URL: [Electric Duet · player under the GPL](#)

Alex Patalenski — improved Electric Duet player `src/plugins/duet.s` transcribes his 1989 speaker player, published by Emil Dotchevski, instruction for instruction: its 73-cycle loop is kept in an aligned segment so the timing stays his. Apple II DeskTop uses the same player. URL: [the listing and byte code · Apple II DeskTop](#)

Cybernesto — electric-mock (GPL v3) The Mockingboard rendition of Electric Duet songs in `src/plugins/duet.c` follows his player: one AY tone per voice. A2FileCmd times it with the VIA instead of delay loops, and scales the periods to the speaker player. URL: [electric-mock repository](#)

Andy McFadden — CiderPress II The format notes of CiderPress II guided the MacPaint and AppleWorks data base and spreadsheet readers, and its test files are the real samples the tests use. The ones A2FileCmd reads ship in the XL image's DEM0/CIDERPRESS folder (`data/CP2/README.TXT` lists them). URL: [CiderPress II · format notes](#)

POM2 — Arnaud Verhille Separate emulator project used for repeatable Apple IIe, //c and disk-device verification. URL: [POM2 repository](#)

ADTPro Virtual-drive and disk-transfer workflows support moving the supplied .dsk images to real hardware. URL: [ADTPro project](#)

Reading the source notices

The repository keeps attribution next to the affected implementation:

- `src/crt0.s` and `src/crt0_loader.s` identify the cc65 startup source and describe the local staging changes.
- `src/vsdrive.s` identifies the a2tools file and explains the adapted serial entry points.
- `src/format.c` and `src/format_diskii.s` identify the Hyper-FORMAT lineage and separate the original formatter work from A2FileCmd integration.
- `src/unshrink.s` describes the documented archive algorithms; its decoder is an independent implementation written for the overlay memory limits.
- `src/plugins/duet.s` names the Electric Duet players it transcribes and what was changed (zero page, alignment, exits); `src/plugins/duet.c` names the Mockingboard rendition it follows.

The project does not copy Norton Commander, A2Command, ProDOS or ADTPro executables. Their interfaces, manuals and protocols are references or inspirations. The generated demo files are also original: `tools/mkdemo.py` creates the pictures, music, text and sample archives during the build. When redistributing a modified build, keep this section, the source notices and LICENSE with the program so the attribution remains visible.

Reference index

For readers who want to compare the implementation with its references:

- A2FileCmd source and issue tracker
- cc65 documentation
- cc65 Apple II library sources
- a2tools source and releases
- ADTPro documentation
- ADTPro source

- ProDOS 8 technical reference
- Electric Duet, Paul Lutus
- Alex Patalenski's Electric Duet player
- electric-mock, Cybernesto
- Apple II DeskTop file types
- CiderPress II format notes
- NuLib library and Binary II references
- Apple II FAQ archive
- A2Command archive mirror
- Norton Commander archive
- POM2 emulator
- ZX-Art AY / PT3 catalogue
- ZX-Art PT3 API
- ABSTRACT (Ra, 1999) PT3
- dh2020rt (EA, 2020) PT3
- Music (VAD, 2002) PT3
- Old Skool For Demodulation (EA, 2020) PT3
- realtime blast (EA, 2026) PT3
- Yazzie: final theme (nq, 2019) PT3
- ana_ng.pt3 in dos33fsprogs
- mA2E_3.pt3 in dos33fsprogs
- Vortex Tracker II
- zxtunes.com author list
- zxtunes.com Macros archive
- zxtunes.com Korund archive
- Vince Weaver pt3_lib

Historical archives may move or disappear; the repository copies the relevant attribution and source-path information so the record remains useful if a mirror changes.

Only the fragments identified above are derived from external code or documented routines. The A2FileCmd overlays, ProDOS walkers, image readers, tests, demo data and user interface were written for this project. Generated demo files are created by `tools/mkdemo.py`; they are not copied from the inspiration projects. See the repository source comments and LICENSE for the complete copyright and redistribution terms.